FPGA Tactical Controller for StarCraft II Micro Scenarios

EECS 3216 — Updated proposal draft, November 12, 2026

Target: Tier A, subject to instructor approval; existing demo provisionally approved by TA

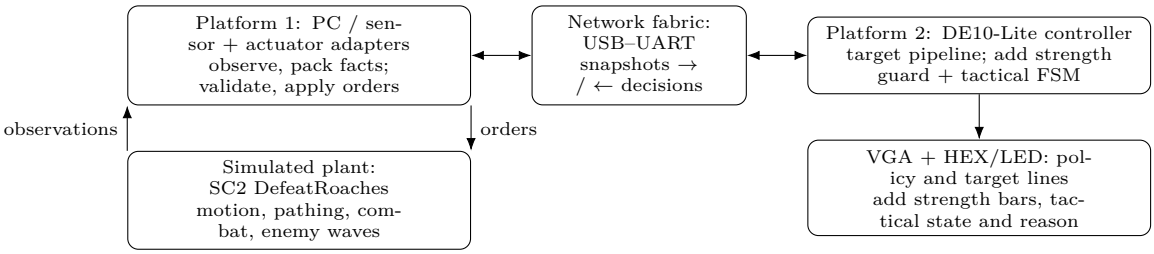
Group: [Lead name/ID], [Development partner name/ID], [Report partner name/ID], [Demo partner name/ID]

1. Project goal and existing demo

Our working DE10-Lite controller selects attack targets in StarCraft II (SC2) *DefeatRoaches*: Marines fight enemy Roach waves. A switch selects nearest-target or focus-fire tactics; VGA shows units and target lines.

The lead reports a defeat-to-victory demo after changing tactics. We will extend PR #6 with strength assessment and retreat, making two decisions visible: *whom to attack, and when to disengage.*

2. Feedback architecture: a simulated cyber-physical plant



Adapted from Lee–Seshia Fig. 1.1 [1]. API observations act as sensors and API orders as actuators. SC2 is the simulated plant: it executes movement and combat. The PC transports facts and applies the FPGA's decisions.

3. Tactics and visible decisions

- **Focus fire — whom to attack [2].** Target the weakest enemy in range, otherwise the nearest. Finishing an enemy removes a source of damage. Toggle target lines, then compare fights from the same start.
- **Lanchester-inspired strength — whether to engage [3].** Estimate each side's strength as $P = kN^2$, using living-unit count N and a fixed matchup coefficient k reflecting damage rate and opponent health. A small square lookup and integer comparisons suit FPGA logic. VGA adds friendly/enemy strength bars.
- **Tactical FSM — which orders to issue.** ATTACK when engagement is permitted; RETREAT when strength or health is low; HOLD at a defined safe point. Display the state and reason; latch retreat until reset.

The strength model is an approximation; range, wounds and waves affect outcomes. Judge retreat by units preserved. Keep the scope to 16 units/side and this matchup; defer kiting, damage reservations, mixed armies and general navigation.

4. Why FPGA, and how we will demonstrate it

- **Digital-system design:** custom UART at 115200 baud, record registers, a three-stage distance/ranking pipeline, interacting FSMs and 640×480 VGA on the MAX 10 at 50 MHz.
- **Visible hardware value:** communication, decisions and video run concurrently. Inspect one frozen snapshot, toggle the policy and show the resulting targets and core cycles. Demonstrate a rejected bad packet and recovery.
- **Evaluation:** check outputs and transitions against an independent reference; verify Quartus timing and resources. Report outcomes, core cycles and communication latency; no CPU speedup is assumed.

5. Team responsibilities and delivery

- **Project lead:** existing implementation; architecture, algorithms/FSM, host integration and final review.
- **Development partner:** independent board/UART/VGA setup; strength/status display and testing assistance.
- **Report partner:** report structure, references, architecture explanation and results presentation.
- **Demo partner:** scenario setup, equipment checklist, evidence capture, narration and rehearsal.

Retreat integration: Nov. 20. Feature freeze: Nov. 26. Rehearsal: Nov. 27–Dec. 1. Demo/submission: Dec. 2.

[1] E. A. Lee and S. A. Seshia, *Introduction to Embedded Systems*, 2nd ed., Fig. 1.1, p. 6; Sec. 3.3 (2017).

[2] D. Churchill and M. Buro, *Portfolio Greedy Search and Simulation for Large-Scale Combat in StarCraft*, Sec. III (2013).

[3] M. Stanescu, N. Barriga and M. Buro, *Using Lanchester Attrition Laws for Combat Prediction in StarCraft*, AIIDE (2015). Our guard is an engineering adaptation, not their learned model.